



What should we expect before we simulate?

Order-of-magnitude physics of a 1.4 m plastic scintillator bar for the SHiP Timing Detector — energy deposition, total internal reflection, photon transport, reflector losses and timing observables, derived before any Geant4 output is shown.

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Napkin first. Geant4 second.



THE OBJECT

One bar is a cell of a 50 m² detector

TOP · up to 70 SiPM along the bar



END · 8 SiPM per face

1400 mm × 60 mm × 10 mm

0.084 m²

geometric area of one bar

~600

bars needed for a 50 m² detector

9.6k – 50k+

SiPM in total, for 16 / 70 / 86 per bar

DESIGN QUESTION

Which combination of optical transport, SiPM number and position, and time estimator gives the best resolution per readout channel — knowing that every bar is one small cell of an almost 50 m² surface?

BEST TESTED SO FAR — NOT A PROVEN OPTIMUM

EJ-230 + END + Vikuiti with air gap:

$\sigma(T0) \approx 58.6$ ps, $\sigma x \approx 10.3$ mm, $v_{eff} \approx 184.6$ mm/ns, $N_{pe} \approx 1400$ at the centre (no electronic jitter).

A small loss in resolution against a large reduction in channels may be globally preferable.



The simulation must not cover our understanding

Geant4 returns a number. Only a prior analytic estimate tells us whether that number is physics or a bug.

1 — NAPKIN

Snell, geometry, exponential attenuation and order statistics fix the order of magnitude and, more importantly, the direction of every trend: which material gives more light, which gives better timing, which face imposes more bounces.

2 — GEANT4

Confirms the scale, adds the distributions the napkin cannot produce — Landau fluctuations, path-length dispersion, angular mixing — and quantifies what we already expected qualitatively.

Knoll, Ch. 8, on optical Monte Carlo: these codes are accurate only to the extent that the assumptions about surface reflection, bulk absorption and the refractive indices of every medium are correct. Those are exactly the three inputs a napkin calculation can police.

If the two disagree, one of them is wrong — and finding out which one is the actual result.



1

STEP 1 — ENERGY DEPOSITION

How much energy does one muon leave?

$$dE/dx \approx 2 \text{ MeV/cm} \text{ (minimum ionizing, plastic)}$$

$$T = 10 \text{ mm} \rightarrow E_{\text{dep}} \approx 2 \text{ MeV}$$

- One number, one line of arithmetic — but it sets the scale of everything downstream.
- It is deliberately crude: it fixes an order of magnitude, not a distribution.
- Every later factor in this talk multiplies this 2 MeV.

$\approx 2 \text{ MeV}$

deposited in 10 mm of scintillator

WHAT THE NAPKIN CANNOT GIVE

Geant4 adds the event-by-event fluctuation:
full Bethe–Bloch, Landau tail, δ -ray production.
The mean survives; the width is simulation territory.



2

STEP 2 — SCINTILLATION YIELD

From 2 MeV to twenty thousand photons

$$N_{\gamma} = E_{\text{dep}} \cdot Y \approx 2 \times 9700 \approx 1.94 \times 10^4 \text{ photons}$$

$$\approx 2 \times 10^4$$

optical photons per crossing muon

Optical properties used in the simulations

Material	Yield [ph/MeV]	τ_d [ns]	λ_{att} [cm]
EJ-200	10000	2.1	380
EJ-204	10400	1.8	160
EJ-230	9700	1.5	120

THE TRADE-OFF IS ALREADY VISIBLE

The three yields are the same to within 7 %. What separates the materials is not how much light they make, but how much survives 70 cm of travel (λ_{att}) and how fast the emission decays (τ_d). Those two pull in opposite directions.



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STEP 3 — TOTAL INTERNAL REFLECTION

How much of that light can stay inside?

$$ns \sin \theta_c = na, \quad ns \approx 1.58 \rightarrow \theta_c = 39.3^\circ$$

$$\alpha_c = 90^\circ - \theta_c = 50.7^\circ \text{ measured from the surface}$$

$$|u_\perp| < \cos \theta_c = \sqrt{(1 - 1/ns^2)} = 0.774$$

An exact trigonometric identity combined with Snell — no Taylor expansion, no small-angle approximation.

$$\text{isotropic: } P(|u_y| < a) = a = 0.774 \text{ per wall family}$$

$$\text{both transverse dimensions: } f_{TIR} = 2a - 1 \approx 0.548$$

KNOLL EQ. 8.15 — SLAB GEOMETRY

$F = \sqrt{1 - (n_1/n_0)^2} = \cos \theta_c$. The textbook quotes 75 % at $n = 1.5$; at $n = 1.58$ it returns 0.774 — our single wall family, exactly.

BUT THE BAR IS NOT A SLAB

Two wall families must hold simultaneously: $f_{TIR} = 2a - 1 = 0.548$, i.e. ≈ 0.274 towards each end. That split matters later.

AND IT IS NOT A CYLINDER EITHER

Knoll Eq. 8.14, meridional rays: $\frac{1}{2}(1 - n_1/n_0) = 18.4\%$ per direction. A different geometrical quantity from the 80 % quoted for EJ-228 — not comparable as it stands.



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STEP 4 — LONGITUDINAL VELOCITY

How fast does the light travel — and how axial is it?

$$v_g \approx c/ns \approx 190 \text{ mm/ns} \rightarrow 700 \text{ mm in } \approx 3.7 \text{ ns (axial ray)}$$
$$v_x = (c/ns) \cos \beta \rightarrow \text{diagonal rays are longitudinally slower}$$

Reading Geant4 backwards: what equivalent angle does the measured propagation slope imply?

$$\text{END: } \cos \beta_{\text{eff}} = 184.6 / 189.9 \rightarrow \beta_{\text{eff}} \approx 13.5^\circ$$
$$\text{TOP: } v_{\text{eff}} \approx 145 \text{ mm/ns} \rightarrow \beta_{\text{eff}} \approx 40^\circ$$

KEY IDEA

END and TOP do not merely read the light in different places — they select different angular families of photons. END is dominated by an almost axial, early population; TOP integrates markedly more diagonal trajectories.

READ BEFF AS A SLOPE, NOT AS A BEAM

v_{eff} is a statistical slope built from the earliest photons, so β_{eff} is the equivalent axial angle of that slope — not evidence of a mono-angular population. Knoll also notes that adding escape surfaces lowers the mean number of reflections.



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STEP 5 — BOUNCE COUNTING

Geometry, not the reflector, sets the number of bounces

$$N_{\text{bounce}} \approx d \cdot \tan \beta / D \quad (\text{geometric estimate})$$

2D limiting case, for a ray just outside TIR on that wall family: $\beta \approx 50.7^\circ \rightarrow \tan \beta \approx 1.22$, travelling $d = 700$ mm from the centre.

$D = 10$ mm

≈ 85

bounces, thin face

$D = 60$ mm

≈ 14

bounces, wide face

THE TWO TRANSVERSE DIMENSIONS ARE NOT EQUIVALENT

A $1400 \times 60 \times 10$ mm bar is thin in one direction and wide in the other. The same photon direction produces six times more reflections against the 10 mm face than against the 60 mm face.

$$N_y \approx d|u_y| / (W|u_x|)$$

$$N_z \approx d|u_z| / (T|u_x|)$$

The 3D form is the rigorous one: the integer bounce count also depends on where in the cross-section the photon starts.



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STEP 6 — REFLECTOR LOSSES

A 2 % loss per bounce becomes an exponential attenuation

$$PR = R^N = \exp(N \ln R), \quad N \approx d \tan \beta / D$$
$$PR(d) = \exp(-d / \Lambda_{\text{refl}}), \quad \Lambda_{\text{refl}} = -D / (\tan \beta \cdot \ln R)$$

 $\Lambda_{\text{refl}} (\beta = 50.7^\circ)$ **$\approx 0.4 \text{ m}$**

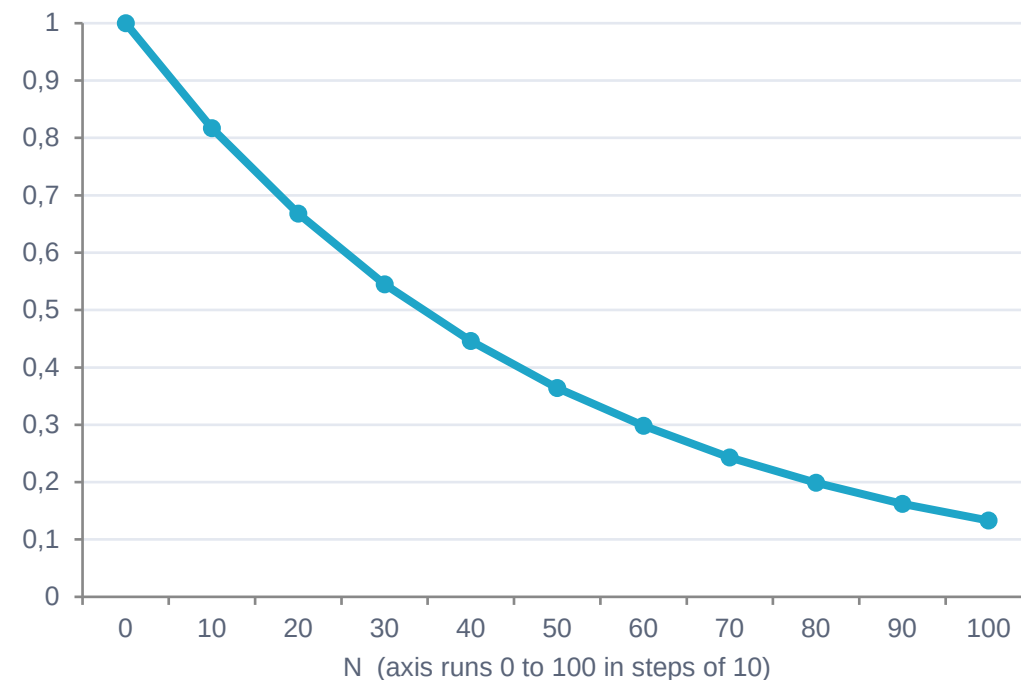
10 mm face

 $\Lambda_{\text{refl}} (\beta = 50.7^\circ)$ **$\approx 2.4 \text{ m}$**

60 mm face

These are not attenuation lengths of Vikuiti. Reflector attenuation is reflectivity $\times$ geometry $\times$ photon direction, and both numbers are conditioned on the angle chosen above.

Survival vs number of reflections N (R = 0.98)



$$0.98^1 = 0.98 \quad \cdot \quad 0.98^{10} = 0.82 \quad \cdot \quad 0.98^{50} = 0.36 \quad \cdot \quad 0.98^{100} = 0.13$$



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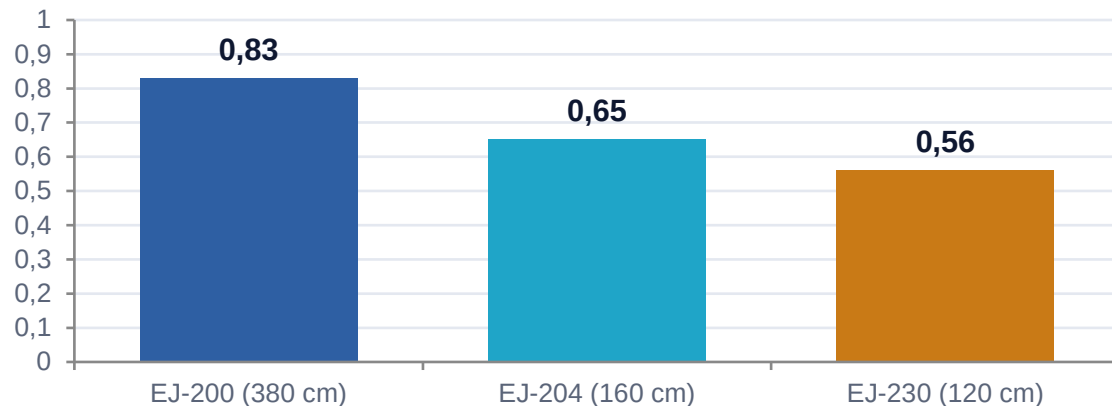
STEP 7 — MATERIAL CHOICE

Bulk attenuation and decay time pull in opposite directions

$$s = d / |ux| \rightarrow P_{\text{bulk}}(ux) = \exp(-d / (\lambda_{\text{att}} |ux|))$$

The familiar $\exp(-70/\lambda_{\text{att}})$ at the bar centre is the axial best case, $|ux| = 1$. Every diagonal ray travels further and survives less.

Axial best case, centre to one end



NAPKIN PREDICTION, MADE BEFORE THE RUN

EJ-200 must deliver the most light to the ends (largest λ_{att}).
EJ-230 must lose the most light, but its faster kinetics ($\tau_d = 1.5$ ns vs 2.1 ns) should win on timing.

GEANT4

Npe: EJ-200 > EJ-204 > EJ-230

σ_t : EJ-230 < EJ-204 < EJ-200

Both orderings reproduced, in the predicted directions — the curves are on the next slide.



The ordering was predicted, then measured

$N_{pe}(x) \cdot EJ-200 / EJ-204 / EJ-230$

Insert the light-yield-versus-position curves from the END + Vikuiti campaign.
The napkin predicts the ordering $EJ-200 > EJ-204 > EJ-230$ at every x , and a steeper fall for the shorter λ_{att} .

$\sigma(T0)(x) \cdot EJ-200 / EJ-204 / EJ-230$

Insert the timing-resolution-versus-position curves from the same campaign. The napkin predicts the opposite ordering, $EJ-230 < EJ-204 < EJ-200$, driven by τ_d .

Two figures, two orderings, opposite directions — both written down before the Monte Carlo was run. This is the difference between illustrating a simulation and validating a prediction.



Photon budget: multiply the factors and check the scale

$$1.94 \times 10^4$$

photons produced

$$\times 0.274$$

TIR-guided towards ONE end (0.548
/ 2)

$$\times 0.56$$

bulk attenuation, 70 cm

$$\times 0.48$$

end coverage, 288 / 600 mm²

$$\approx 1.4 \times 10^3$$

photons at that end

$$\times PDE \approx 0.4 \rightarrow \approx 5.7 \times 10^2 \text{ PE per end} \approx 1.1 \times 10^3 \text{ for } L + R$$

Geant4, EJ-230 + END + Vikuiti at the centre: $N_{pe} \approx 1400$. The chain closes, and the excess is what the reflector-recovered population should add.

The factors are not strictly independent (coverage, PDE, incidence angle and multiple chances to hit a sensor are correlated). This is a scale check, not a prediction.

COUNT PER END, ALWAYS

Half the guided population goes each way. Mixing a both-ends fraction with a one-end coverage silently doubles the answer.

THE OUTSTANDING ANOMALY

A historical END + TIR-only run returned of order 1 PE per event. Before that is called physics, it has to survive the most pessimistic napkin we can write.



Even the worst case leaves two orders of magnitude

Readout-face treatment	Angular acceptance towards one end	PE per end	Implied Vikuiti / TIR-only
Index-matched coupling	0.274 (TIR-guided)	$\approx 5.7 \times 10^2$	≈ 1.2
Air gap, escape cone only	$0.113 = (1 - 0.774) / 2$	$\approx 2.4 \times 10^2$	≈ 3
Reported run	—	$\approx 1 \text{ PE / event}$	$\approx 10^3$

THE ESCAPE CONE CANNOT BE THE EXPLANATION

Knoll warns that TIR must be minimised on the face you read from. But that cone sits at the readout face, so it cuts the Vikuiti configuration by the same factor — it cannot generate a ratio.

The only genuine difference is the side walls: 0.548 guided, versus 0.548 plus whatever the reflector recovers.

THE FALSIFIABLE PREDICTION

Run END + TIR-only and END + Vikuiti under identical conditions and compare per-end yields. The napkin brackets the ratio between roughly 1 and 3 — every other factor cancels.

A ratio near 10^3 is not a physical result. It is a surface, boundary or RINDEX definition to be found.



From arrival times to physical observables

$$tL = t0 + (L/2 + x)/veff + \epsilon L$$

$$tR = t0 + (L/2 - x)/veff + \epsilon R$$

SUM

$$T0 = (tL + tR)/2$$

x cancels → vertex time

DIFFERENCE

$$\Delta t = tR - tL = -2x/veff$$

→ longitudinal position

$$\sigma x = (veff / 2) \cdot \sigma \Delta t$$

CONSISTENCY CHECK, NOT AN EXTRA MEASUREMENT

$veff = 184.6 \text{ mm/ns}$ and $\sigma \Delta t = 111.5 \text{ ps}$ give

$$\sigma x = 92.3 \times 0.1115 \approx 10.3 \text{ mm},$$

exactly what the simulation reports. The position resolution is not independent information — it is the timing resolution seen through the propagation velocity.

DECOMPOSITION OF THE MEASURED TIME

$$t_{meas} = t0 + t_{scint} + t_{prop} + t_{sensor} + t_{elec}$$

t_{prop} is a distribution of path lengths, not a constant. Knoll flags exactly this case: for fast emitters of large dimensions, the spread in photon path lengths can limit the achievable timing resolution.



Napkin versus Geant4 and the literature

Quantity	Napkin prediction	Geant4 / literature	Verdict
Photons per muon	$\approx 2 \times 10^4$	same order	consistent
Trapped fraction, one wall family	0.774	Knoll Eq. 8.15: 0.774	textbook match
Longitudinal velocity	≤ 190 mm/ns, axial limit	184.6 mm/ns $\rightarrow \beta_{\text{eff}} \approx 13.5^\circ$	consistent
Npe at centre, END + Vikuiti	$\approx 1.1 \times 10^3$ (L + R)	≈ 1400	consistent
Light ordering	EJ-200 > EJ-204 > EJ-230	same ordering	confirmed
Timing ordering	EJ-230 < EJ-204 < EJ-200	same ordering	confirmed
σ_x from $\sigma\Delta t$	10.3 mm	10.3 mm	internal check
Cylinder trapping (EJ-228)	18.4 % per direction (Eq. 8.14)	80 % quoted	different definition — check
Vikuiti / TIR-only yield ratio	between 1 and 3	$\approx 10^3$	AUDIT THE RUN

Seven agreements. One quantity that needs its definition checked. One genuine anomaly.



Two photon populations, two kinds of information



WHAT THE REFLECTOR BUYS

A large increase in total collected light, concentrated in the late part of the arrival distribution.

WHAT IT DOES NOT BUY

A proportional improvement in the statistics of the earliest photons, which is what a leading-edge estimator uses.

Total $N_{pe} \neq$ timing information carried by the first photons



Specular or diffuse? The napkin gives us a falsifiable prediction

SPECULAR — VIKUITI 3M ESR

Preserves the angle of incidence, so a photon outside the acceptance of the readout face is never redirected into it. It also preserves short paths, keeping the dispersion of t_{prop} small.

DIFFUSE — TEFLON, TiO_2 , MGO

Lambert's law, $dI/d\Omega \propto \cos \psi$: the outgoing direction is essentially independent of the incoming one. Every bounce is a fresh chance to be redirected towards a readout window, at the cost of a broader path-length distribution.

Napkin hypothesis: at comparable reflectivity, diffuse scattering should raise the probability of redirecting photons into a readout window, and should broaden t_{prop} . One campaign, one variable, prediction on record.

Stated as a hypothesis, not a conclusion: a real diffuse wrapping differs from a specular one in spectral reflectivity, absorption and surface finish, not only in angular law. Knoll reports diffuse reflectors generally collect light better — but that comparison is made on light yield, not on timing. Kilvington's data (Knoll Fig. 8.17) show TIR with a reflective covering sitting systematically above TIR alone, both falling with guide length.



Why assume the first photon is optimal?

$$TO^{(1)} = (tL^{(1)} + tR^{(1)}) / 2 \rightarrow \sigma \approx 58.6 \text{ ps}$$

EVIDENCE THAT $k = 1$ NEED NOT BE THE OPTIMUM

In an independent EJ-228 cylinder study $\sigma(k)$ is non-monotonic, with an optimum near $k \approx 8$. The number does not transfer to a 1.4 m bar — the principle does.

How much timing information are we throwing away by using only $t^{(1)}$?

ORDER STATISTICS

$t^{(k)}$ for $k = 1 \dots 100$; fractional rank $k \approx f \cdot N_{pe}$

MULTI-PHOTON ESTIMATORS

mean / median / trimmed mean of the first m ; $\sum w_i t_i$;
accumulated-PE threshold; digital CFD; temporal template

MULTI-READOUT ESTIMATORS

BLUE(L, R) instead of a plain average; END + TOP through the residual covariance

optical $\rightarrow$ SiPM SPTR $\rightarrow$ front-end $\rightarrow$ clock / common mode. 58.6 ps is the optical limit alone: if SPTR is of order 100 ps, winning 14 ps optically may matter less than halving the channel count.



What the napkin tells us to do next

A

Exploit the existing data first

Audit the ROOT files for arrival time, SiPM ID, face, PE, wavelength and true muon position — and fix the per-end versus L+R convention before any ratio is quoted.

B

Close the 1 PE question with a ratio

Re-run END + TIR-only and END + Vikuiti under identical conditions. The napkin brackets the per-end yield ratio between 1 and 3; anything near 10^3 is a surface or RINDEX definition problem.

C

Separate the two ablations

Masking a channel offline is not the same as removing the sensor from the geometry: physical removal changes the optical boundary condition and therefore the transport.

D

Report a Pareto front, not a minimum

σ (mean and worst case in x) against the number of channels, with SPTR and front-end jitter injected. Quadrature does not automatically apply to multi-channel estimators.



Three messages

1

Geometry explains the scale.

TIR, propagation and repeated reflection already predict the dominant trends — and reproduce the textbook where the textbook applies.

2

More light is not necessarily better timing.

Early TIR photons and late reflector-recovered photons carry different information. Total Npe is the wrong figure of merit for a timing detector.

3

The optimisation problem is still open.

The optimum is not the maximum number of SiPM. It is the Pareto optimum of σ_t against the number of channels, once SPTR and front-end jitter are in.

EJ-230 + END + Vikuiti is the best tested point — not yet the optimum.